

Auditing Temporal and Identity Dependence in SCONE-bench: Dual-Provider Historical State Reconstruction of 417 Smart-Contract Incidents

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Abstract

Context: Historical smart-contract benchmarks are often counted by incident rows, although a row’s usable pre-incident history and independence depend on deployment time, code reuse, and proxy architecture. **Objective:** We audit these dependencies in the current 417-case SCONE-bench cohort without evaluating a detector or making population-level vulnerability claims. **Method:** We reuse closed historical envelopes whose upstream evidence records verified status, a strict snapshot closure, block-hash-pinned reads, and two provider families. For each case, we measure whether the contract existed 1 hour, 24 hours, 7 days, and 30 days before its incident; compare chain-address, address-only, exact-runtime, and metadata-stripped runtime identities; and count specified ERC-1967 implementation/beacon and ERC-1167 indicators. **Results:** All 417 cases existed at least 24 hours before their incident, 352 (84.4%, Wilson 95% interval 80.6–87.6%) existed at 7 days, and 275 (65.9%, 61.3–70.3%) existed at 30 days. Address identity assigns 14 rows to duplicate groups, versus 46 under exact runtime and 56 after metadata stripping. Sixty-seven cases (16.1%, 12.9–19.9%) expose at least one specified proxy indicator. **Conclusion:** Evaluation capacity and effective independence in this cohort vary with the temporal landmark and identity abstraction. The audit provides a reproducible cohort profile and boundary conditions, not evidence of detector accuracy, causal risk, or prevalence beyond SCONE-bench.

Keywords: smart contracts; benchmark validity; historical state; code reuse; proxy contracts; reproducibility

1 Introduction

Smart-contract security evaluation increasingly uses curated datasets and executable historical incidents. Frameworks such as SmartBugs standardize static-analysis execution (di Angelo et al., 2023); SC-Bench provides a large audit-oriented corpus (Xia et al., 2025); EVMbench and later reevaluations expose the sensitivity of agent conclusions to task construction, scaffolds, and possible contamination (Wang et al., 2026; Peng et al., 2026). CyberChainBench extends historical-fork evaluation across chains (Huang et al., 2026). These resources make previously difficult experiments possible, but a benchmark’s nominal row count is not automatically its number of temporally usable or independent evaluation units.

The current SCONE-bench repository contains 417 historical incidents (Anthropic, 2026). An earlier report described a 405-case snapshot and local historical forks (Anthropic, 2025). We treat the newer 417-case cohort as a finite benchmark population and ask three deliberately narrow questions:

1. At which pre-incident landmarks did each benchmark contract already exist?
2. How does effective dependence change across address- and bytecode-based identity abstractions?
3. How many rows expose a specified, observable proxy-linkage indicator?

Our contribution is the joint audit and its reproducible artifacts. The temporal test itself, hashing, metadata stripping, and proxy standards are established mechanisms; we make no “first” claim. The useful negative result is equally important: the audit cannot establish detector performance, vulnerability prevalence, causal risk, or real-world effectiveness.

Table 1: Strongest-prior-art matrix. “Not located” means not found within the documented search scope, not universal absence.

Work	Historical incidents	Historical state	Identity audit	Temporal audit	Proxy audit
SCONE-bench	Yes	Fork block	Not located	Not located	Not located
Re-evaluating EVMbench	Yes	Evaluation dependent	Contamination focus	No	No
CyberChain-Bench	Yes	Historical forks	Not located	Not located	Proxy subset for patching
DIVE	No	Lifecycle features	Duplicates removed	No	Not central
This audit	417 rows	Dual-provider, block-hash-pinned upstream evidence	Four abstractions	1h/24h/7d/30d	Specified observable indicators

2 Related work and strongest prior art

Dataset validity is already a central concern in smart-contract research. DIVE explicitly reports lifecycle-specific features and duplicate removal (Alsunaidi et al., 2026); prior empirical work finds extensive source/function-level cloning in Ethereum contracts (Khan et al., 2023). These studies motivate identity sensitivity but do not answer it for SCONE-bench’s incident rows. EIP-1898 standardizes block-hash selectors for canonical historical reads (Ethereum Improvement Proposals, 2019a); ERC-1967 defines proxy implementation, beacon, and admin slots (Ethereum Improvement Proposals, 2019b); ERC-1167 defines recognizable minimal-proxy runtime code (Ethereum Improvement Proposals, 2018). Table 1 records the bounded prior-art comparison used to avoid overstating novelty.

3 Materials and methods

3.1 Frozen cohort and evidence gate

The unit of analysis is one of the 417 case envelopes in the frozen local snapshot. The analysis fails closed unless every envelope is marked `VERIFIED`, has `strict_snapshot_closed=true`, and records both Alchemy and Infura provider families. It reuses upstream deployment time, incident time, runtime bytecode, and normalized proxy fields. The source tree is read-only; all derivatives are written to an isolated MPP directory.

The cohort contains 226 BSC, 181 Ethereum, 9 Base, and 1 Arbitrum cases (Table 2). This is a benchmark census, not a probability sample of deployed contracts.

Table 2: Cohort composition (T3).

Chain	Cases	Percent
BSC	226	54.2
Ethereum	181	43.4
Base	9	2.2
Arbitrum	1	0.2

3.2 Estimands

For case i , let d_i be deployment time and t_i incident time. Landmark eligibility is $E_{i,\ell} = \mathbb{K}[t_i - d_i \geq \ell]$ for $\ell \in \{1\text{h}, 24\text{h}, 7\text{d}, 30\text{d}\}$. This definition was frozen after an integration test detected and corrected an early implementation that had incorrectly substituted snapshot-cutoff lead for deployment age. The correction and its timing are preserved in the deviation record.

Four identity keys are compared: chain–address, address only, SHA-256 of exact runtime bytecode, and SHA-256 after removing a recognized Solidity CBOR metadata trailer. Each identity reports unique

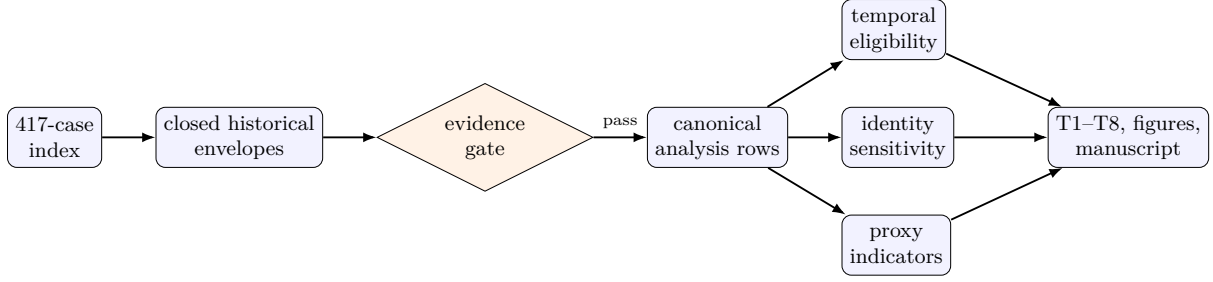


Figure 1: Analysis architecture. Arrows represent data flow, not causality. Editable Mermaid source is included with the artifact.

keys, groups of size at least two, rows in such groups, maximum group size, cross-address groups, and cross-chain groups. Equality defines a fingerprint family only.

Observable proxy linkage is the union of non-null ERC-1967 implementation, non-null ERC-1967 beacon, and recognized ERC-1167 target fields. The admin slot is reported separately because it does not identify a delegate target by itself. This deliberately incomplete rule produces a lower-bound indicator count.

3.3 Statistical and reproducibility plan

Because the frozen cohort is fully enumerated, the primary estimands are counts and proportions. Two-sided 95% Wilson intervals are supplied as descriptive uncertainty aids. No hypothesis test, multiplicity adjustment, imputation, model fitting, or causal estimator is used. All computations use deterministic Python standard-library code, sorted output, fixed CSV schemas, and no network access or randomness. Algorithms 1–3 give the auditable logic.

Algorithm 1: Temporal landmark eligibility. For each case i , parse deployment time d_i and incident time t_i . Fail closed if either value is absent or if $d_i > t_i$. For each predeclared landmark $\ell \in \{1\text{h}, 24\text{h}, 7\text{d}, 30\text{d}\}$, set $E_{i,\ell} = \mathbb{1}[t_i - d_i \geq \ell]$. Report $\sum_i E_{i,\ell}/N$ with a two-sided 95% Wilson interval. The historical snapshot cutoff is audited separately and never substituted for deployment age.

Algorithm 2: Identity sensitivity. For every case, construct four fingerprints: chain–address, address only, SHA-256 of exact runtime bytecode, and SHA-256 of runtime bytecode after removing the recognized Solidity CBOR metadata trailer. Group equal fingerprints, count unique identities, groups of size at least two, rows in those groups, cross-address groups, and cross-chain groups. No bytecode equality result is interpreted as semantic equivalence.

Algorithm 3: Observable proxy linkage. Normalize the upstream ERC-1967 implementation, admin, and beacon values and the recognized ERC-1167 target. Count an implementation, beacon, or ERC-1167 target as an observable proxy indicator; report admin separately because an admin slot alone does not establish a proxy target. The resulting count is a lower bound, not a complete proxy classifier.

4 Results

4.1 Temporal eligibility

All 417 contracts existed at both the 1-hour and 24-hour landmarks. At 7 days, 352 were eligible (84.4%, Wilson 95% interval 80.6–87.6%); at 30 days, 275 were eligible (65.9%, 61.3–70.3%). Deployment-to-incident age ranged from 25.48 hours to 55,430.26 hours, with median 2,325.71 hours. Figure 2 and Table 3 report the prespecified outcomes.

4.2 Identity sensitivity

Chain–address and address-only keys each yield 410 unique identities, seven duplicate groups, and 14 rows in those groups. No additional cross-chain same-address collision appears in this cohort. Exact runtime bytecode yields 388 identities and 46 duplicate-group rows; ten of its 17 duplicate groups span addresses, and three groups (16 rows) span chains. Metadata stripping yields 382 identities and 56

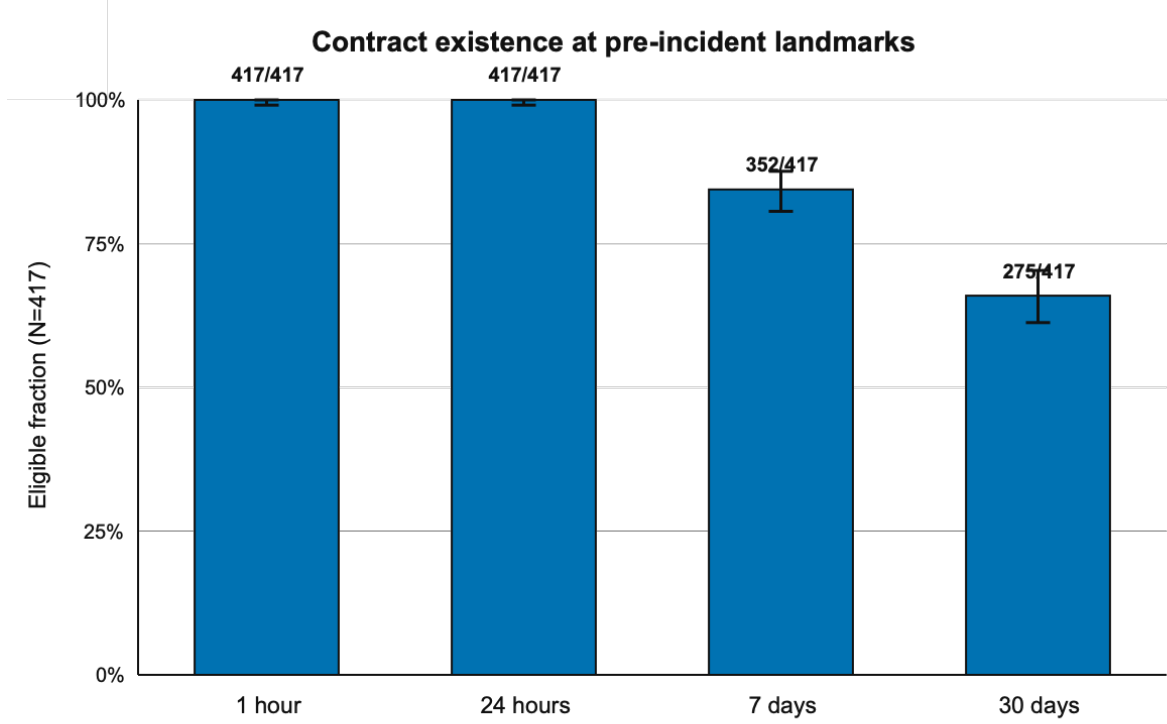


Figure 2: Contract existence at pre-incident landmarks. Error bars are two-sided 95% Wilson intervals; counts use the fixed denominator $N = 417$.

duplicate-group rows; 14 of 21 groups span addresses and three groups (17 rows) span chains. The maximum runtime-family size is nine (Table 4, Figure 4).

4.3 Observable proxy linkage

Fifty-four cases carry a non-null implementation slot, one a non-null beacon slot, and 12 a recognized ERC-1167 target. Their union contains 67 cases (16.1%, Wilson 12.9–19.9%). Thirty-eight admin-slot observations are reported separately. Overlap means component counts cannot be summed. The result is a cohort-specific lower bound under the declared indicator rule.

5 Robustness and boundary analyses

Metadata stripping adds ten duplicate-group rows and ten cross-address duplicate rows relative to exact runtime identity; it adds one cross-chain row. This demonstrates that an apparently small preprocessing choice changes the effective dependence estimate. Conversely, chain–address and address-only results coincide in this cohort. These are abstraction-sensitivity results, not proof that one identity is universally correct.

The temporal denominator is also landmark-dependent. A study requiring a 30-day pre-incident window has 275 eligible units before any additional exclusion, not 417. A study requiring only 24 hours retains all rows. This paper does not evaluate whether a particular downstream task actually needs each landmark.

6 Discussion

The main implication is operational: a historical incident benchmark should publish a landmark-conditioned denominator and at least one dependence sensitivity analysis. A single nominal row count hides both. The results do not imply that 30 days is preferable to 7 days, nor that metadata-stripped bytecode is the definitive unit. They show how those choices alter the usable cohort.

Table 3: Primary temporal and proxy results (T4).

Section	Metric	n	N	Percent	Wilson 95% interval
Temporal	1 hour	417	417	100.0	99.1–100.0
Temporal	24 hours	417	417	100.0	99.1–100.0
Temporal	7 days	352	417	84.4	80.6–87.6
Temporal	30 days	275	417	65.9	61.3–70.3
Proxy	implementation slot	54	417	12.9	10.1–16.5
Proxy	admin slot (separate)	38	417	9.1	6.7–12.3
Proxy	beacon slot	1	417	0.2	0.0–1.3
Proxy	ERC-1167 target	12	417	2.9	1.7–5.0
Proxy	any specified indicator	67	417	16.1	12.9–19.9

Table 4: Identity-abstraction mechanism audit (T5).

Identity	Unique	Groups	Rows	Cross-address groups	Cross-chain groups	Max size
Chain-address	410	7	14	0	0	2
Address only	410	7	14	0	0	2
Exact runtime	388	17	46	10	3	9
Metadata-stripped runtime	382	21	56	14	3	9

Proxy architecture creates an additional linkage layer. The 67-case lower-bound indicator count signals that contract address alone can obscure implementation reuse or indirection. Future work could resolve implementation graphs at each landmark, identify semantic clone families, and test how cluster-aware resampling changes detector or agent uncertainty. Those are new studies, not results of this audit.

6.1 Threats to validity

Construct validity. Deployment and incident timestamps, bytecode, and proxy fields are reused from upstream envelopes. Exact bytecode is not semantic behavior, and the proxy rule omits nonstandard and diamond-style patterns. **Internal validity.** The script validates declared closure fields but does not independently recreate every upstream provider response. Correlated upstream or provider errors may remain. **External validity.** SCONE-bench is a curated incident benchmark, not a probability sample; no percentage generalizes to all contracts, chains, vulnerabilities, or incidents. **Conclusion validity.** Wilson intervals are descriptive aids for a fixed cohort, not a basis for population inference. **Novelty validity.** The documented search found related benchmark-validity, lifecycle, and clone studies; no independent search challenger has certified the remaining differentiator.

7 Reproducibility, ethics, and responsible use

The bundle includes a row-level provenance manifest, input manifest, analysis code, tests, result CSV/JSON files, editable T1–T8 tables, SVG masters, PNG derivatives, Mermaid diagram sources, LaTeX source, BibTeX, and mechanical QA outputs. No participant recruitment, intervention, private key use, or new chain scraping occurs. The analysis processes public smart-contract incident metadata, but downstream row-level redistribution rights require human/legal review before public release. Security-sensitive details are not expanded beyond the upstream benchmark. The study makes no exploit-generation or operational safety claim.

Generative AI assisted planning, code generation, drafting, and internal checking under human direction. It is not an author, scientific verifier, or accountable submitter. A human author must validate every result, disclosure, citation, and source-rights statement before submission.

8 Conclusion

Within the current 417-case SCONE-bench cohort, all contracts support a 24-hour pre-incident landmark, while 352 support 7 days and 275 support 30 days. Dependence rises from 14 duplicate-group rows under address identities to 46 under exact runtime and 56 after metadata stripping. Sixty-seven cases expose at least one specified proxy indicator. These findings support reporting landmark-conditioned

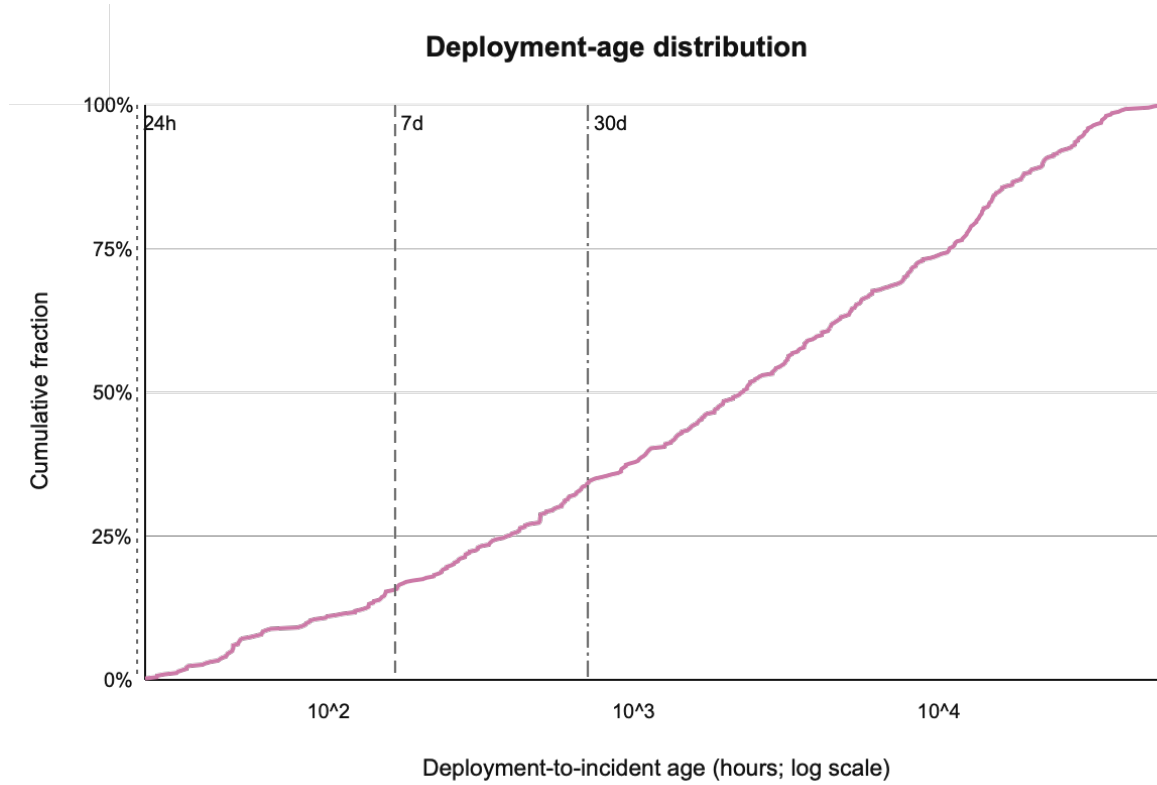


Figure 3: Empirical cumulative distribution of deployment-to-incident age on a logarithmic hour scale. Vertical lines mark 24 hours, 7 days, and 30 days.

Table 5: Robustness and boundary comparisons (T6).

Comparison	Baseline	Alternative	Difference and boundary
Address duplicate rows	14	14	0; no added cross-chain address collision
Exact vs stripped duplicate rows	46	56	+10; metadata changes identity
Exact vs stripped cross-address rows	32	42	+10; semantic near-clones still excluded
Exact vs stripped cross-chain rows	16	17	+1; state/configuration may still differ

and identity-conditioned benchmark denominators. They do not establish detector performance, causal vulnerability risk, or ecosystem prevalence. The artifact is mechanically reproducible locally; independent reproduction and accountable human approval remain prerequisites to upload.

Statements and Declarations

Funding: Pending accountable-author confirmation.

Competing interests: Pending accountable-author confirmation.

Ethics approval and consent: No human or animal participants and no intervention; final institutional determination, if required, is pending accountable-author confirmation.

Data availability: The current SCONE-bench repository is publicly accessible. Redistribution of this paper’s row-level derivatives is pending source-rights review; aggregate tables and code are prepared in the local bundle.

Code availability: Analysis code and tests are prepared in the local reproducibility bundle; public repository/DOI pending author approval.

Author contributions: Pending named human authors and CRediT confirmation.

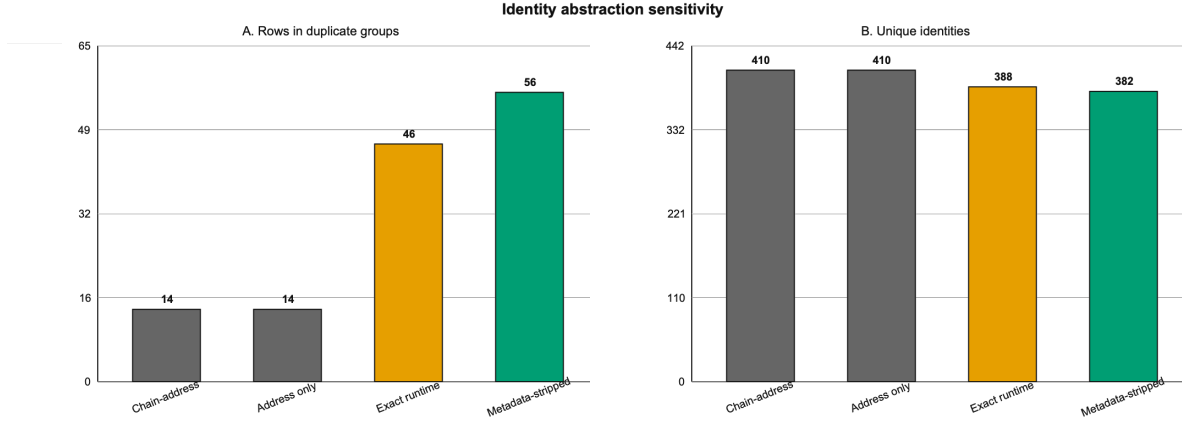


Figure 4: Sensitivity to the identity abstraction. Direct labels make the result interpretable without color.

AI-use disclosure: Generative AI assisted planning, code generation, drafting, and internal checking. Human authors must take responsibility for the final work.

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A Minimum artifact table set

Table 6: Proposed audit versus simpler baselines (T2).

Approach	Temporal	Identity	Proxy	Main failure
Row count	No	No	No	Overstates usable independent units
Address dedup	No	Address	No	Misses cross-address families
Exact runtime	No	Exact code	Partial	Metadata and proxy layers remain
This audit	Four landmarks	Four abstractions	Specified indicators	Still a lower bound

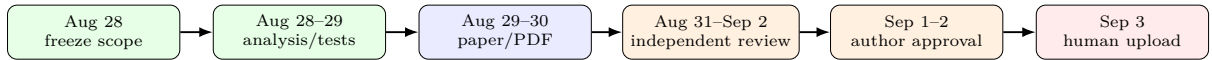
Table 7: Feasibility and authority gates (T7).

Dimension	Status	Evidence	Remaining gate
New collection	N/A	Read-only 417-case reuse	None
Determinism	Verified mechanically	Standard-library tests	Independent reproduction
Rights	At risk	SCONE Apache-2.0	Row-level release review
Scientific independence	Blocked	Root produced/self-reviewed	Qualified independent reviewer
Submission authority	Blocked	Author/declarations unresolved	Accountable human approval

Table 8: Negative findings and scope controls (T8).

Finding	Effect on claim
No controls or detector evaluation	Cohort-validity audit only
Proxy rule incomplete	67 is a lower-bound indicator count
Bytecode equality is not semantics	Groups are fingerprint families only
Cohort is not a probability sample	No ecosystem generalization
Novelty not independently certified	Submission remains gated

B Execution timeline



The dates after local paper/PDF completion are targets, not completed scientific or authority gates.